A Case Study of Classroom Podcast in Ohlone College

A

Dissertation Proposal

Presented to the

Faculty of the Graduate School of Education

Alliant International University

by

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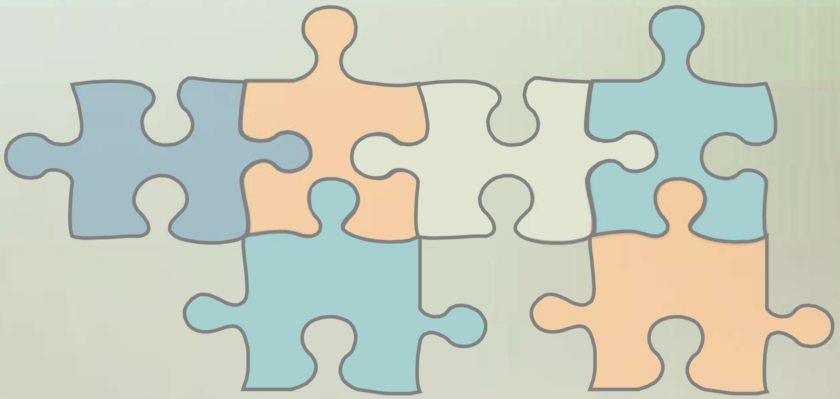
Organization of the Dissertation

- Chapter 1 - Background and Rationale
- Chapter 2 - Literature Review
- Chapter 3 - Methodology
- Chapter 4 - Results
- Chapter 5 – Conclusions

Proposal



Definition of Terms

- ***MP3***: MPEG-1 is a digital audio encoding file format used to store digital audio podcasts for playback on a portable media player or MP3 player.
- ***Ohlone's iTunesU***: A podcasting service that allows educators at Ohlone College to publish lectures and related class information through audio podcast and allows students to subscribe and download podcasts automatically via iTunes.
- ***Portable digital media device***: Small electronic digital handheld device, such as a MP3 player smart cell phone, PDA or computer.
- ***Podcast:*** A prerecorded classroom lecture and learning materials produced as audio and video files.
- ***Podcasting***: A process of creating and distributing classroom learning materials through audio and video podcasts to the Internet, allowing users to receive files automatically.

Background

- Since the 1990s Internet and Web technologies have created opportunities for exchange information, technology-facilitated teaching and learning, as well as distance learning for higher education.
- The Internet and related Web technologies have expanded educational offerings from classroom instruction to Web-based courses and now through the use of mobile and portable devices.
- Mobile, Internet, and Web technologies have seen significant growth in the past decade, with rapid growth in use by the teenage population.
- In recent years, advancements in mobile and portable technologies have revolutionized the way educators and students communicate, interact, and engage in classroom learning activities.

Current Technology Trends

Popular Mobile Devices

What is Podcasting?

- Audio/or video (or multimedia) file on the Internet
- Subscription-enabled through RSS (Really Simple Syndication)
- Practically playback on any mobile devices, MP3 (iPod) devices or personal computers

Why Podcasting?

- The idea of podcasting represents new thinking in the delivery of educational content for playback and/or viewing lectures and lessons on student personal media players.
- Podcasting is essentially a recording of class lectures and reviewing them outside the classroom.
- This approach is not a new concept, but podcasting of class lectures via mobile devices for students at the community college is a new concept.

Statement of the Problem

- According to the Pew Internet and American Life Project (2005), almost 9 in 10 teens were Internet users and approximately 11 million teens went online daily. About 87% of teens in the United States used the Internet—about 21 million youth, up from about 17 million in late 2000.

- The Pew Internet and American Life Project (2009) shows that mobile phone use has climbed steadily among teens ages 12 to 17 – to 63% in fall of 2006 to 71% in early 2008.

Purpose of the Study

The purpose of this study is to investigate the use of classroom podcasts as supplemental course materials for playback of lectures and other lessons and the relationship of this use with the test scores Ohlone College's students.

- The study will be examined the effects of classroom podcasts and the use of mobile and portable media devices, including iPods and MP3/MP4 players, and the impact of these devices on student learning outcomes via classroom instruction with supplemental audio course materials.

- The prediction is that students who listen to classroom podcasts are more likely to achieve higher test scores than students who do not listen to classroom podcasts.

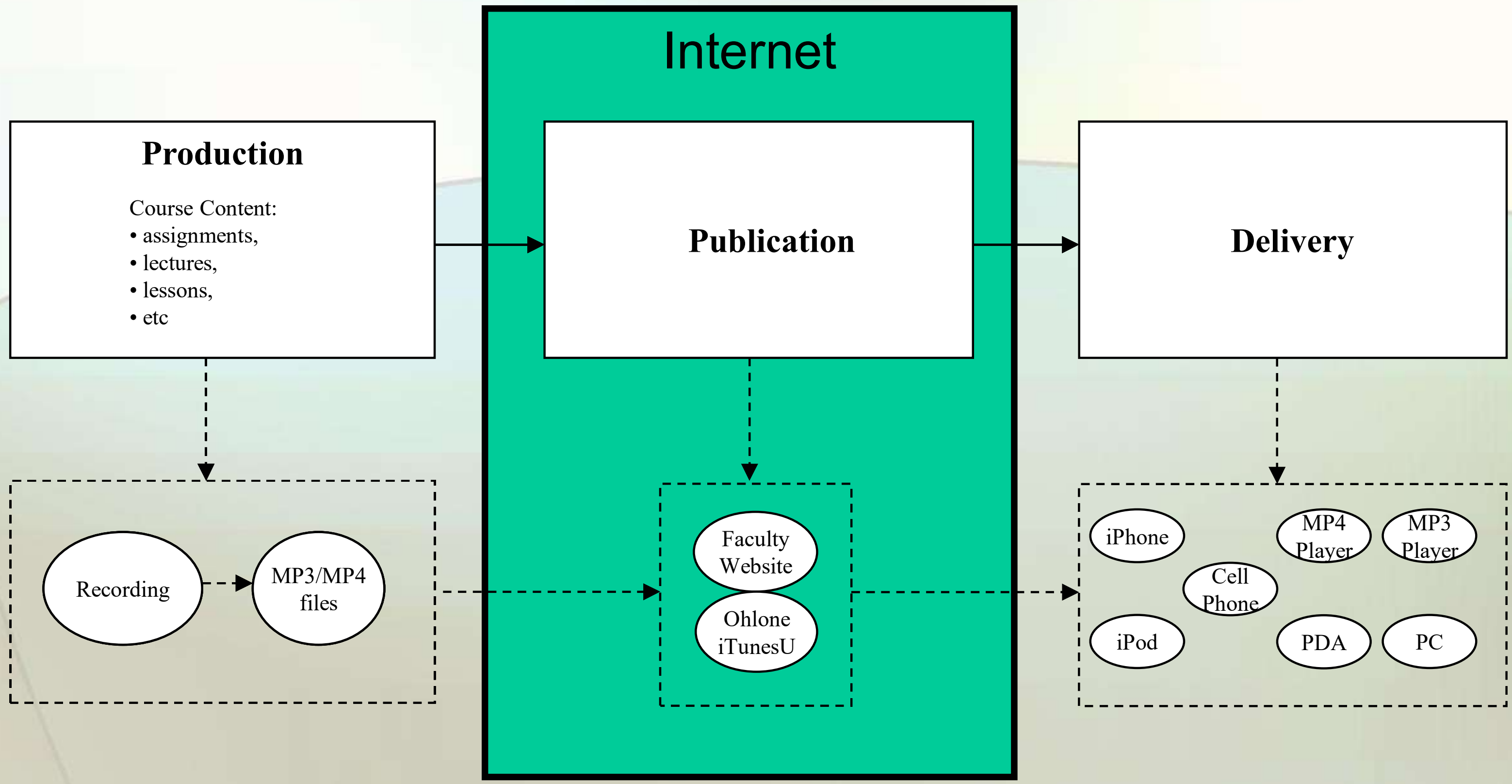
Research Questions

1. Are there differences between two types of educational instructions (traditional classroom instruction and classroom instruction supplemented with podcasts) with regard to the students' test scores?
2. Is there a significant difference between the number of hours students listen to classroom podcasts (1 hour or less, 2 hours, 3 hours or more, and not at all) with regard to their test scores?
3. Is there a significant relationship between the number of hours students listen to classroom podcasts (1 hour or less, 2 hours, 3 hours or more, and not at all) and the average class attendance (less than other classes, same as other classes, and more than other classes)?

Research Question (cont.)

4. Is there a significant difference between genders with regard to the number of hours students listen to classroom podcasts (1 hour or less, 2 hours, 3 hours or more, and not at all)?
5. Is there a significant difference between age groups with regard to the number of hours students listen to classroom podcasts (1 hour or less, 2 hours, 3 hours or more, and not at all)?

Classroom Podcasting



Ohlone iTunesU

Ohlone Community College District
43600 Mission Boulevard
Fremont, CA 94539-0390

Ohlone College Newark
Center for Health Science &
39399 Cherry Street Newark, CA 94560

www.ohlone.edu

About Ohlone College

- Ohlone College is a part of the Ohlone Community College District, with campuses in Fremont and Newark.
- To the south of Ohlone College is the Silicon Valley, best known for the silicon chip innovators and the headquarters of such leading high-technology manufacturers as Apple Computer, Ebay, Google, Hewlett Packard, Intel, Oracle, Sun, Yahoo, and many more.

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Ohlone Community College District - Community Demographics

Race	Fremont	%	Newark	%	Union City	%	District Total	%
American Indian	692	0.3%	29	0.1%	18	0.0%	739	0.2%
African American	6,012	2.9%	1,015	2.5%	4,195	6.0%	11,222	3.5%
Asian	84,657	40.6%	7,271	17.7%	20,919	30.0%	112,847	35.3%
Filipino	11,387	5.5%	3,058	7.4%	12,844	18.4%	27,989	8.8%
Hispanic	32,108	15.4%	14,656	35.6%	15,225	21.8%	61,989	19.4%
Pacific Islander	2,097	1.0%	564	1.4%	510	70.0%	3,171	1.0%
White	65,566	31.5%	13,223	32.2%	14,489	20.8%	93,278	29.2%
Other	5,936	2.8%	1,309	3.2%	1,548	2.2%	8,793	2.8%
Total	208,455		41,125		69,748		319,328	

Source: Ohlone College. (2009). *Ohlone data source: Institutional Research Office*

A World of Cultures United in Learning

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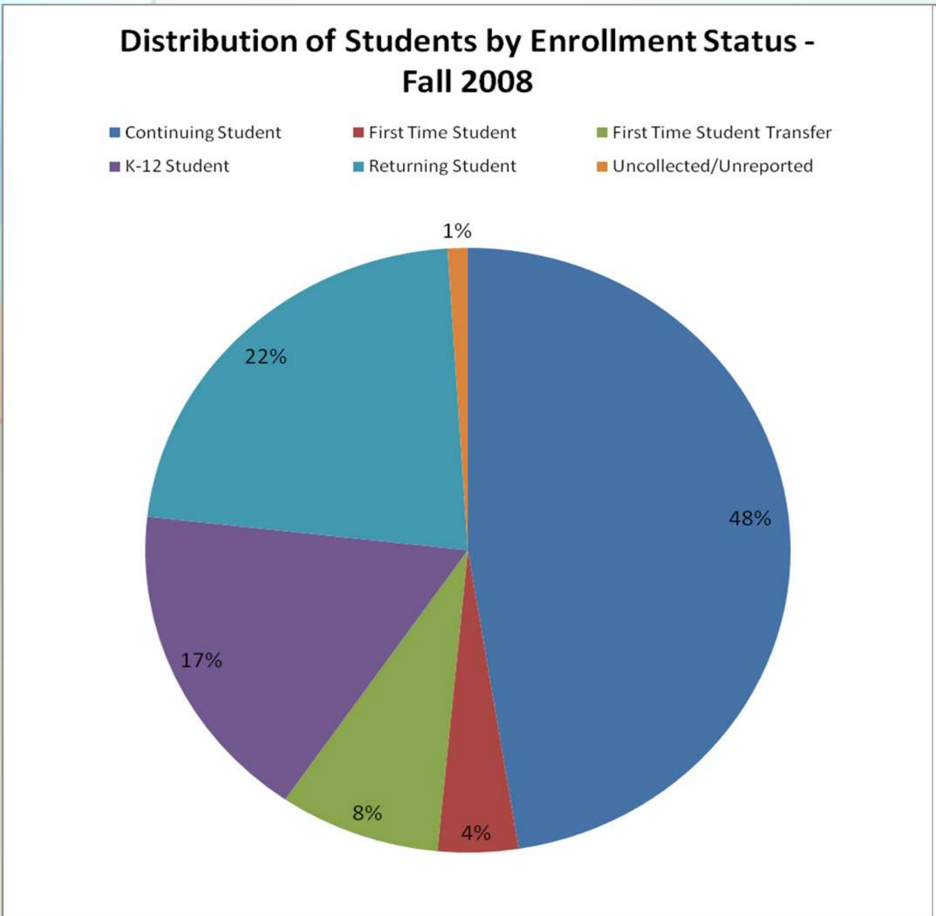
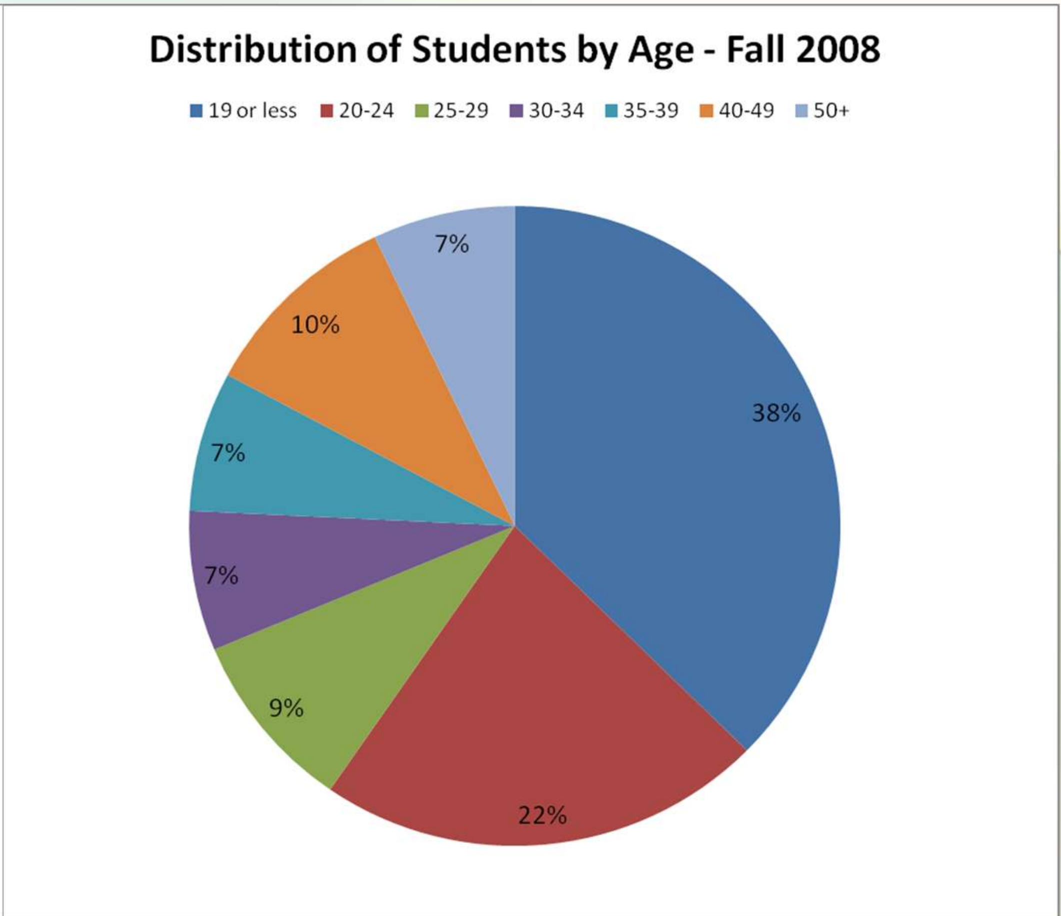




Source: Ohlone College. (2009). *Ohlone data source: Institutional Research Office*

Quasi-Experimental Design

Posttest

Nonequivalent Control Group Design	X_1 (midterm)	O	X_1 (final exam)	O
	X_2 (midterm)	O	X_2 (final exam)	O

Repeated Posttest

Figure 1: Grouping of participants. X1 = classroom podcasting (experimental treatment), X2 = no classroom podcasting (control treatment), O = posttest.

Source of Invalidity

Source of Invalidity										
	Internal								External	
	History	Maturation	Testing	Instrumentation	Regression	Selection	Mortality	Selection Interactions	Pretest-X Interaction	Multiple-X Interference
Nonequivalent Control Group Design	✓	✓	-	-	✗	✓	✗	✓	-	-

Grouping of Participants

Table 1

Participants Grouped by Method of Classroom Instruction

Group	Type of instruction	Designation
1	Traditional classroom instruction	Control group
2	Classroom instruction supplemented with classroom podcast	Treatment group

Treatment Group

- During the Fall 2009 semester, all students from lecture-based course, History of the United States (HIST-117A) were invited to participate in the study.
- All students received traditional classroom instruction supplemented *with* classroom podcasts.
- The maximum enrollment for History of the United States is ***120*** students at Ohlone College.

Control Group

The same lecture-based course from the previous Spring 2009 semester, History of the United States (HIST-117A), taught by the same instructor in traditional classroom instruction *without* podcasting will be used as a control group.

Data Collection Instrument

A three-part handout questionnaire will be used to collect students' responses regarding classroom podcast usage:

a) demographics questions

b) podcasting questions

c) open-ended questions

Dependent Variables

Table 3

Dependent Variables Used in Data Collection and Analysis

Variable	Type	Assigned value	Scale of measurement
Midterm	Numeric	Number value	Scale
Final Exam	Numeric	Number value	Scale

Categorical Variables

Table 2

Categorical Variables Used in Data Collection and Analysis

Variable	Type	Assigned value	Scale
Age group	Categorical	1 = 19 years and younger 2 = 20 to 24 years 3 = 25 to 39 years 4 = 30 to 34 years 5 = 35 to 39 years 6 = 40 to 49 years 7 = 50 years and older	Ordinal
Class attendance	Categorical	1 = less than other classes 2 = more than other classes 3 = same as other classes	Nominal
Ethnicity	Categorical	1 = African American 2 = Asian 3 = Filipino 4 = Hispanic 5 = Pacific Islander 6 = White 7 = other	Nominal
Gender	Dichotomous	1 = Male 2 = Female	Nominal
Academic major	Categorical	1 = Arts and sciences 2 = Business 3 = Engineering 4 = Other	Nominal

Other Types of Variables

Table 4

Other Types of Variables Used in Data Collection and Analysis

Variable	Type	Assigned value	Scale of measurement
Grade received	Numeric	1 = A 2 = B 3 = C 4 = D 5 = F 6 = other	Scale
Hours per week listen to podcasts	Numeric	Number value	Scale
Attitudes toward classroom podcasting	Numeric	1 = Strongly disagree 2 = Disagree 3 = Agree 4 = Somewhat agree 5 = Strongly agree	Ordinal
Year in school	Numeric	1 = First year 2 = Second year 3 = Third year 4 = Fourth year 5 = Fifth year or higher	Ordinal

SPSS Data Dictionary

*ClassPodcastsData_Fall09.sav [DataSet1] - SPSS Data Editor

File Edit View Data Transform Analyze Graphs Utilities Add-ons Window Help

	Name	…	…	…	Label	Values	…	…	Align	Measure
1	InstructionType	…	2	0	Instruction type	{0, Traditional Instruction}…	…	…	Right	Nominal
2	Age	…	2	0	Age Group	{1, 19 years and younger}…	…	8	Right	Ordinal
3	Attendance	…	2	0	Class Attendance	{1, Less than other classes}…	…	8	Right	Nominal
4	Attitude	…	2	0	Attitude toward …	{1, Strongly disagree}…	…	8	Right	Ordinal
5	CourseGrade	…	3	0	Grade Received	{0, A}…	…	8	Right	Scale
6	Ethnic	…	2	0	Ethnicity	{1, Asian}…	…	8	Right	Nominal
7	Gender	…	2	0	Gender	{1, Female}…	…	8	Right	Nominal
8	Hour	…	2	0	Frequency of u…	None	…	8	Right	Scale
9	Major	…	2	0	Academic Major	{0, Art}…	…	8	Right	Nominal
10	Year	…	2	0	Year In College	{1, First}…	…	8	Right	Ordinal
11	Midterm	…	3	2	Midterm test sc…	None	…	8	Right	Scale
12	Final	…	3	2	Final examinati…	None	…	8	Right	Scale

Data View **Variable View**

SPSS Processor is ready

IRB

- Permission to conduct the study from the Office of Student Services at Ohlone Community College
- Permission to conduct the study from the course instructor
- Informed Assent Letter for Students

Analyzing Data

- Personal computer and SPSS (statistical analysis application) will be used to enter the data, analyze, compute the statistics, and interpret the results.
- Descriptive and inferential statistics will be used

Descriptive Statistics

<u>Central Tendency:</u>

- Mean
- Median
- Mode

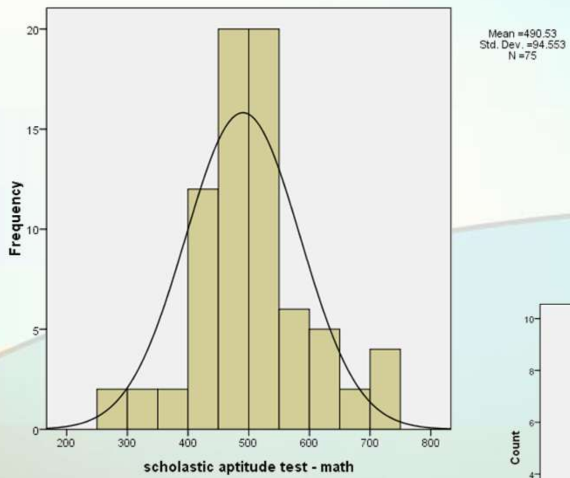
<u>Variability:</u>

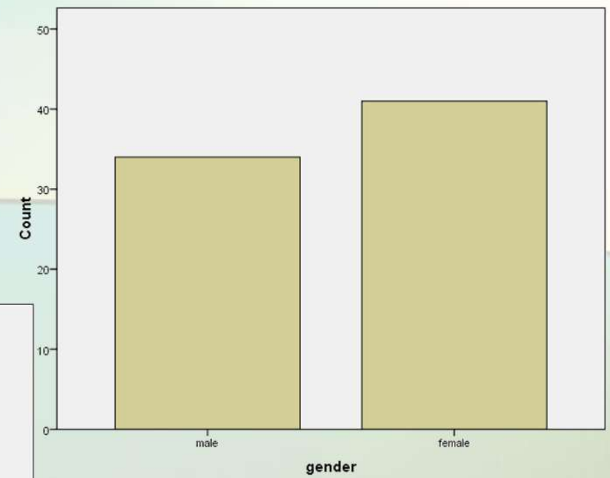
- Range
- Standard Deviation

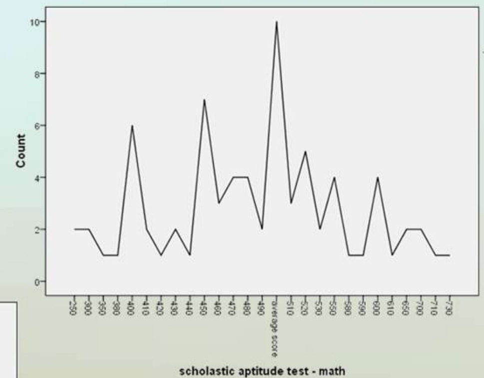
<u>Graphs:</u>

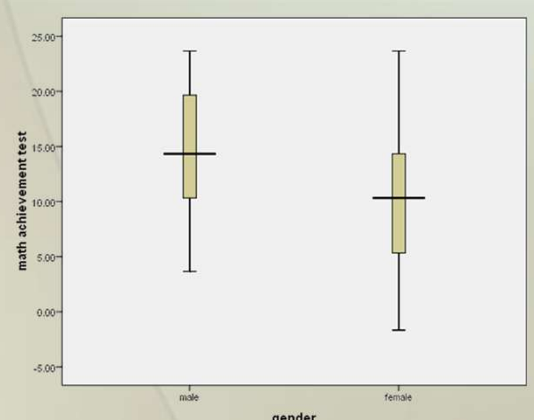
- Frequency Distribution (Nominal, Ordinal, Normal)
- Bar Chart (Nominal, Ordinal, Normal)
- Histogram (Ordinal, Normal)
- Frequency Polygon (Ordinal, Normal)
- Box and Whiskers Plot (Ordinal and Normal)

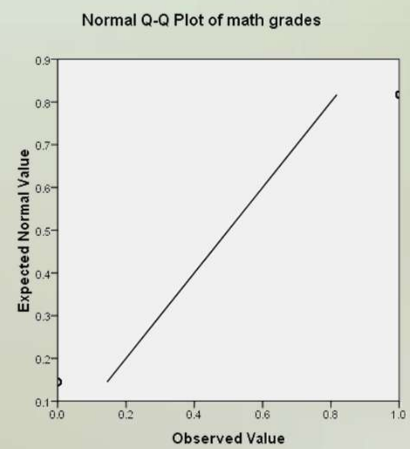
Graphs


Mean =490.53
Std. Dev. =94.553
N =75
Frequency
scholastic aptitude test - math

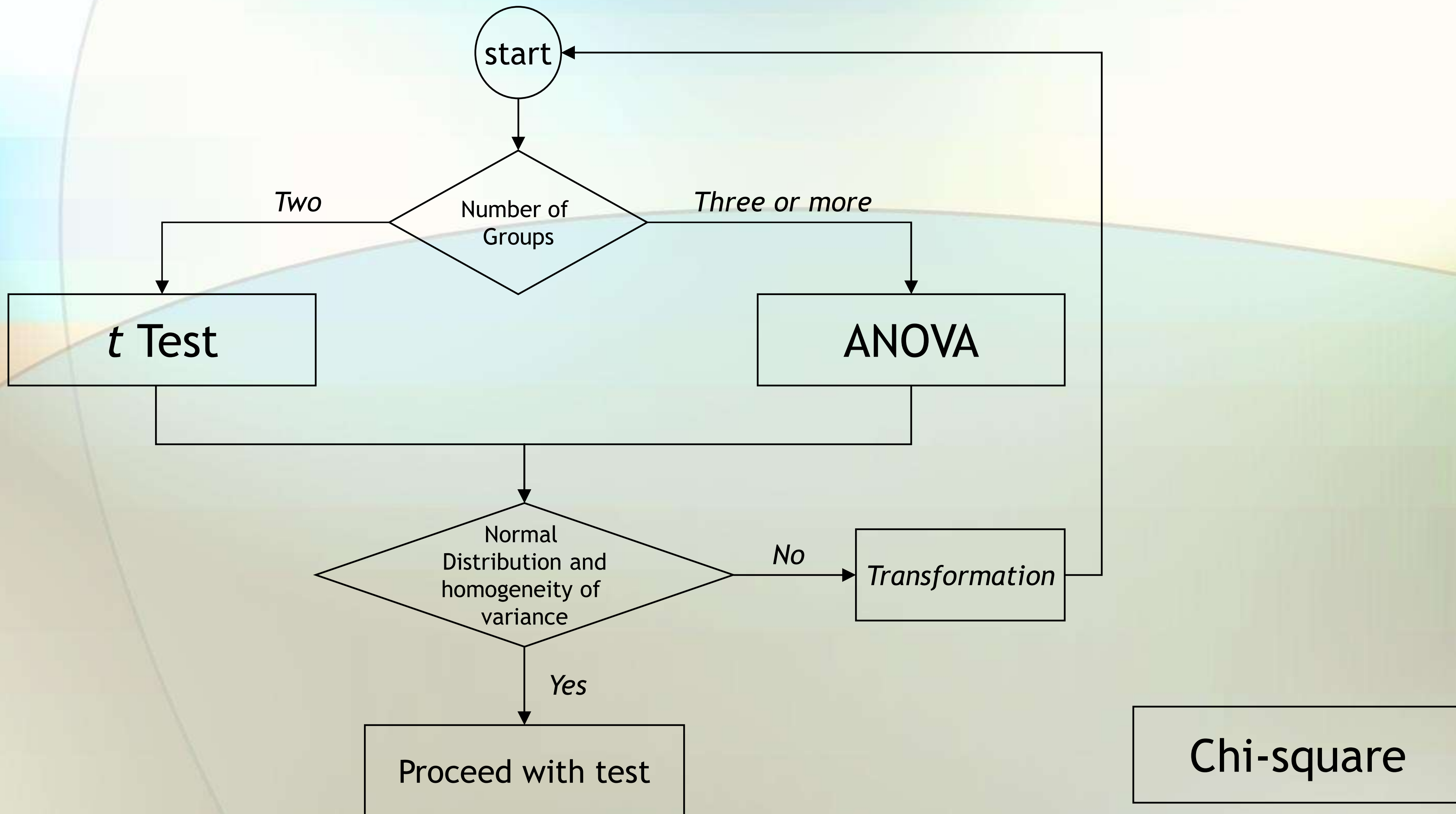

Count
male
female
gender


Count
scholastic aptitude test - math


math achievement test
male
female
gender


Normal Q-Q Plot of math grades
Expected Normal Value
Observed Value

Inferential Statistical Analysis



Independent Samples *t* Test

1 Midterm Posttest

X_1 (midterm)	O	X_1 (final exam)	O
X_2 (midterm)	O	X_2 (final exam)	O

2 Repeated Posttest for final exam

Paired Samples *t* Test

1 Paired *t* Test

X_1 (midterm)	O	X_1 (final exam)	O
X_2 (midterm)	O	X_2 (final exam)	O

2 Paired *t* Test

ANOVA

For three or more categorical independent variables (e.g., age groups, classroom attendances, genders, and number of hour listen to classroom podcasts), a one-way analysis of variance (ANOVA) will be used to measure the differences.

One-Way ANOVA

Table 2

Categorical Variables Used in Data Collection and Analysis

Variable	Type	Assigned value	Scale
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One-Way ANOVA (cont.)

Table 4

Other Types of Variables Used in Data Collection and Analysis

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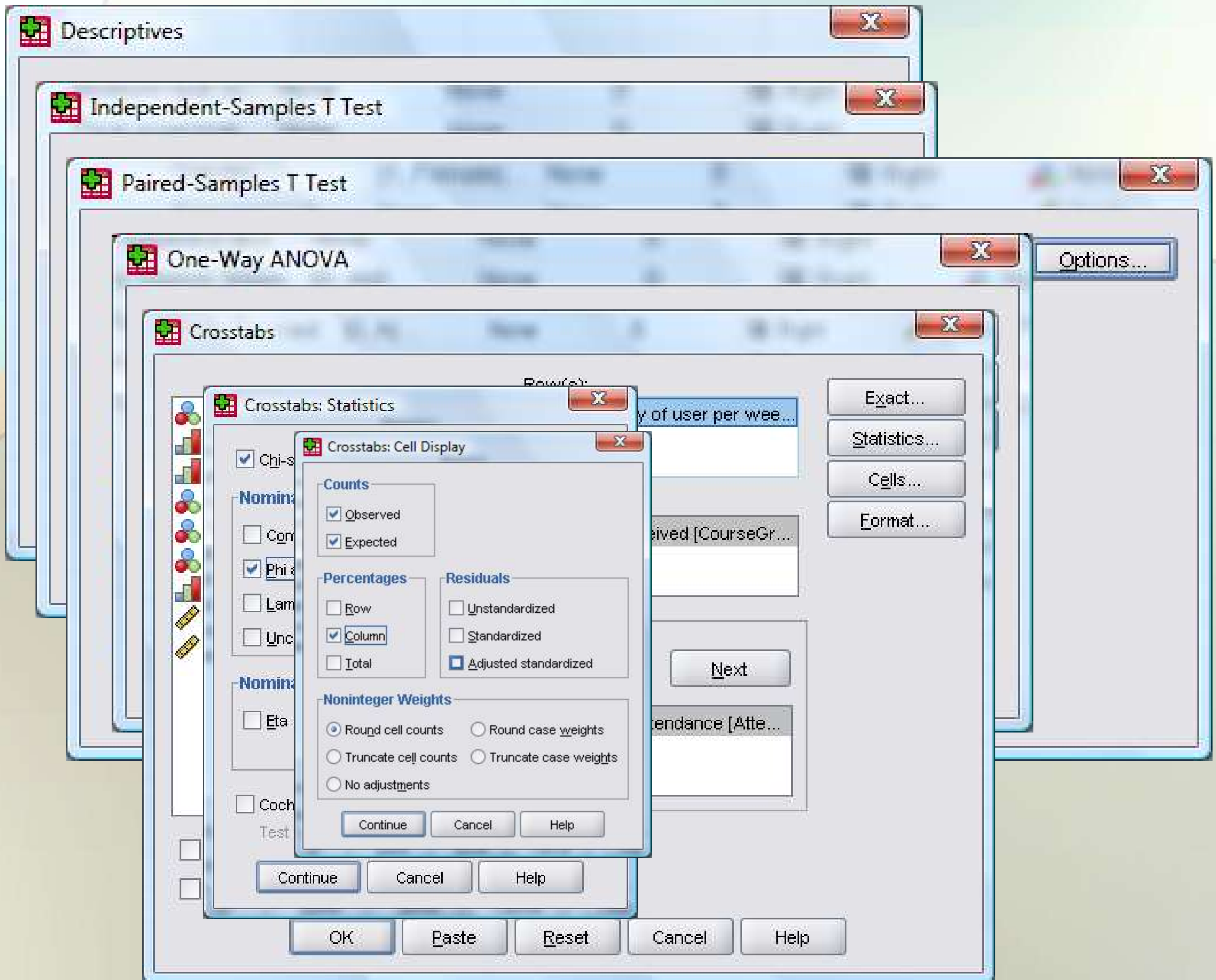
Data Analysis With SPSS



Results and Drawing Conclusions

1. The differences between two types of educational instructions (classroom instruction and classroom instruction supplemented with podcasts).
2. The difference between the number of hours students listen to classroom podcasts with regard to their test scores.
3. The relationship between the number of hours students listen to classroom podcasts and the average class attendance.
4. The difference between genders with regard to the number of hours students listen to classroom podcasts.
5. The difference between age groups with regard to the number of hours students listen to classroom podcasts.

Conclusions

1. The important question for consideration is whether the use of podcasting effectively improves student learning.
2. The findings from this study could be used to determine whether podcasting (production, publication, and delivery of audio and video files) is an instructional path.
3. The findings from this study may help educators to determine the necessary balance between podcast learning and traditional classroom interaction.
4. The findings from this study could be used to expand technology-based learning and provide assistance to educators who desire to develop and share podcasts containing course materials with students.
5. Data gathered in this study could be used to provide leadership and support for future directions of technology-based instruction in higher education.

Summary

This study examines the effects of classroom podcasts and the use of mobile and portable media devices, including iPods and MP3/MP4 players, and the impact of these devices on student learning outcomes via classroom instruction with supplemental audio course materials for community college students.

Q & A

Thank you for making a positive contribution to my education